ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MATHEMATICS

PRACTICE EXAM · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Mathematics practice exam.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give method marks for correct working even if the final answer is wrong (consistent accuracy). Accept other correct methods.

QUESTION 1

1.1 $(2^3)^{2/3} \times (3^3)^{-1/3}$ ✓ $= 4 \times \frac{1}{3}$ ✓ $= \frac{4}{3}$ ✓ (3)

1.2 (a) 5 ✓ (b) $\log(50 \times 2) = \log 100$ ✓ $= 2$ ✓ (3)

1.3 $3^{x+1} = 3^4$ ✓; $x = 3$ ✓ (2)

1.4 $x = \frac{5 \pm \sqrt{(25+32)}}{4}$ ✓✓ $= \frac{5 \pm \sqrt{57}}{4}$; $x = 3{,}14$ ✓ or $x = -0{,}64$ ✓ (4)

1.5 $x \log 2 = \log 50$ ✓; $x = \frac{\log 50}{\log 2}$ ✓ $= 5{,}64$ ✓ (3)

1.6 $\Delta = (-6)^2 - 4(1)(9) = 0$ ✓; roots are real, equal and rational ✓ (2)

1.7 $x + 1 = x^2 - 5$ ✓; $x^2 - x - 6 = 0$ ✓; $(x-3)(x+2) = 0$ ✓; $x = 3$, $y = 4$ ✓; $x = -2$, $y = -1$ ✓ (5)

1.8 $\log x(x-3) = 1$, so $x^2 - 3x = 10$ ✓; $(x-5)(x+2) = 0$ ✓; $x = 5$ ($x = -2$ is rejected because log of a negative number is undefined) ✓ (3)

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QUESTION 2

2.1.1 $x = 0$ ✓; $y = -2$ ✓ (2)

2.1.2 $0 = \frac{6}{x} - 2$ ✓; $x = 3$, i.e. (3 ; 0) ✓ (2)

2.1.3 Asymptotes drawn ✓; correct shape in the 1st and 3rd quadrants relative to the asymptotes ✓; *x*-intercept (3 ; 0) ✓ (3)

2.1.4 $x \in \mathbb{R}$, $x \neq 0$ ✓ (1)

2.1.5 $\frac{6}{-2} - 2$ ✓ $= -5$ ✓ (2)

2.1.6 The graph is shifted 2 units downwards ✓✓ (2)

2.2.1 $g(0) = 1 - 4 = -3$ ✓ (1)

2.2.2 $2^x = 4$ ✓; $x = 2$, i.e. (2 ; 0) ✓ (2)

2.2.3 $y = -4$ ✓; $y > -4$ ✓ (2)

2.2.4 Increasing exponential shape ✓; asymptote $y = -4$ ✓; intercepts (0 ; −3) and (2 ; 0) ✓ (3)

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QUESTION 3

3.1 $A = 25\,000(1 + \frac{0{,}09}{12})^{48}$ ✓✓ $= R35\,785{,}13$ ✓ (3)

3.2 $(1 + \frac{0{,}12}{4})^4 - 1$ ✓✓ $= 0{,}1255 = 12{,}55\%$ ✓ (3)

3.3.1 $A = 180\,000(1 - 0{,}15)^5$ ✓✓ $= R79\,866{,}96$ ✓ (3)

3.3.2 $A = 180\,000(1 - 0{,}15 \times 5)$ ✓ $= R45\,000$ ✓ (2)

3.4 $2P = P(1{,}08)^n$ ✓; $2 = 1{,}08^n$; $n = \frac{\log 2}{\log 1{,}08}$ ✓✓ $= 9{,}01$ years ✓ (4)

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QUESTION 4

4.1 $\frac{1}{2} \times \frac{1}{2} + 1^2$ ✓✓ $= 1{,}25$ ✓ (3)

4.2 $\frac{\sin x \cdot \cos x}{(-\sin x)(-\cos x)}$ ✓✓✓ $= 1$ ✓ (4)

4.3 $LHS = \frac{\sin^2 x}{\sin x \cdot \cos x}$ ✓ $= \frac{\sin x}{\cos x}$ ✓ $= \tan x = RHS$ ✓ (3)

4.4 $\sin x = -0{,}5$ ✓; reference angle 30°, sin negative in quadrants 3 and 4: $x =$ $210°$ ✓ or $x = 330°$ ✓ (3)

4.5.1 $c^2 = 12^2 + 9^2 - 2(12)(9)\cos 50°$ ✓✓ $= 86{,}16$; $c = 9{,}28$ cm ✓ (3)

4.5.2 $\text{Area} = \frac{1}{2}(12)(9)\sin 50°$ ✓ $= 41{,}37$ cm² ✓ (2)

4.5.3 $\frac{\sin B}{9} = \frac{\sin 50°}{9{,}28}$ ✓; $\sin B = 0{,}7429$ ✓; $\hat{B} = 47{,}98°$ ✓ (accept 47,97°) (3)

4.6 Correct cosine shape starting at (0° ; 2) ✓; *x*-intercepts 90° and 270°, minimum (180° ; −2), ends at (360° ; 2) ✓; amplitude 2 ✓; period 360° ✓ (4)

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QUESTION 5

5.1.1 $AB = \sqrt{((4 + 2)^2 + (-5 - 3)^2)}$ ✓ $= \sqrt{100} = 10$ units ✓ (2)

5.1.2 $M(\frac{-2 + 4}{2} ; \frac{3 - 5}{2})$ ✓ $= M(1 ; -1)$ ✓ (2)

5.1.3 $m = \frac{-5 - 3}{4 + 2}$ ✓ $= -\frac{4}{3}$ ✓ (2)

5.1.4 $y - 3 = -\frac{4}{3}(x + 2)$ ✓; $y = -\frac{4}{3}x + \frac{1}{3}$ ✓ (2)

5.1.5 $\frac{3}{4}$ ✓ (1)

5.2.1 $s = \sqrt{(3^2 + 4^2)}$ ✓ $= 5$ cm ✓ (2)

5.2.2 $V = \frac{1}{3}\pi(3)^2(4)$ ✓ $= 37{,}70\ \text{cm}^3$ ✓ (2)

5.2.3 $A = \pi(3)^2 + \pi(3)(5)$ ✓ $= 75{,}40\ \text{cm}^2$ ✓ (2)

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TOTAL: 100

END OF MARKING GUIDELINE